

Questions in Python

1.

Write a Python program to accept the name, age, and city of a person using input. Then, display a sentence like:

`"Hello [Name]! You are [Age] years old and live in [City]."`

Ensure the age input is stored as an integer and perform a type check to confirm it.

2.

Create a program that takes the length, breadth, and height of a box from the user and calculates:

- Volume of the box
 - Surface area of the box
- Also, display the data type of each input and each result.
-

3.

Write a program to input the marks (out of 100) in 5 subjects. Calculate:

- Total marks
- Average marks
- Percentage
- Check and print whether the student passed (all subjects above 33 marks).

Display all results with appropriate formatting and correct data types.

4.

Accept the radius of a circle as input. Calculate and display:

- Diameter
- Circumference
- Area ($\pi = 3.14159$)

Also, round each result to two decimal places and show their data types using `type()`.

5.

Create a Python program that accepts the principal amount, rate of interest, and time (in years) from the user, and calculates the **Simple Interest** using the formula:

$$SI = (P \times R \times T) / 100.$$

Display all input values, intermediate calculations, and the final interest clearly.

6.

Ask the user to enter two numbers and a choice of operation (+, -, *, /). Based on the operation selected, perform the calculation and display the result.

Also, print whether the result is an integer or a float using `isinstance()`.

7.

Write a Python program that asks the user to enter their full name, date of birth (DD/MM/YYYY), and mobile number. Split and display each component clearly.

Example:

Name: Satyender Mishra

Day: 06

Month: 07

Year: 2000

Mobile: 9876543210

8.

Accept a decimal number from the user and convert it to:

- Integer using `int()`
 - Float using `float()`
 - String using `str()`
Print all values and show the type of each converted resu
-

9.

Input a temperature in Celsius and convert it to:

- Fahrenheit: $(C \times 9/5) + 32$
 - Kelvin: $C + 273.15$
Display all results with appropriate labels and round them to 1 decimal place.
-

10.

Take the name, hours worked, and hourly rate as input for an employee. Calculate:

- Gross salary = hours $\times$ rate
 - 10% tax deduction
 - Net salary
Print all values neatly in a formatted summary with clear headings.
-

11.

Create a program that asks the user to enter three sides of a triangle.

- Check whether the input sides form a valid triangle using the triangle inequality.
 - If valid, calculate and print the perimeter and semi-perimeter.
(Use float values for better accuracy.)
-

12.

Write a Python script that accepts the price of 3 items and a discount percentage.

- Calculate the total amount
 - Apply discount
 - Show final bill amount rounded to 2 decimal places
Use proper data types and print the result as a bill summary.
-